

Element F

Functional Consideration	Design Viability Based on Consideration	Is it realistic?	Quotes/Evidence From Source	Relevant Source
Have stimulating reward	Having a light that goes off in each design indicating correct matches of pieces. This works as stimulation for the students and will be positive if the light goes off on correct answers.	Yes	"study found that the use of fluorescent lighting increased the repetitive behaviors of children with autism... noting that fluorescent lighting increased the frequency of stereotypical repetitive behaviors in autistic children."	Theraspecs: https://www.theraspecs.com/blog/light-sensitivity-autism/
Adjustable	The buttons will be attached to the device with magnets or velcro to allow for an easily removable, and therefore adjustable, button design.	Yes	Scheidly discussed alternate ways to secure the buttons to the device and while brainstorming we came up with using some sort of magnetism as it would be sturdy but easier to make and remove.	Eric Scheidly, physics and electronics teacher
Aesthetics	The colors of the device will be bright and appealing to the eyes. We will use reds and oranges to stimulate the students' brains while playing with the device.	Yes	"In general, warm colors elicit happiness and comfort. Bold shades of red, orange, and yellow can stimulate the mind and have an energizing effect on the body."	Handy Handouts: https://www.handyhandouts.com/viewHandout.aspx?hh_number=486&nfp_title=Impact+of+Colors#:~:text=Research%20confirms%20that%20color%20affects,hypersensitive%20or%20bored%20and%20uninterested
Educational value	The use of matching that our design is based around is made to work on repetition. The tiles will correlate to what is being taught in class and this device will allow for an engaging and stimulating repetition to enhance the learning.	Yes	Mrs. Brennan had specifically mentioned using matching to build on the educational content that was being taught during class. "When stimuli are learned by repetition, they are remembered better and retained for a longer time. Studies have shown that the brain forms new pathways	Interview with Brennan and CPD https://cpduk.co.uk/news/importance-of-repetition-in-learning

			when a task is repeated often, thereby optimizing the performance of the skill.”	
Buttons can still be pressed by students with motor disabilities	Some students may have motor disabilities and will not have the precision required to properly hit the buttons. By making them larger and spacing them out more there is a larger margin of error that can be accounted for.	Yes	“Make sure that the control is big enough to operate. Make sure that controls are sufficiently spaced apart. Make sure that controls are positioned to minimize impact of error. Make sure that controls have sufficient color contrast. Make sure that content can be operated using simple inputs on a wide range of pointing devices. Design interfaces to minimize accidental or erroneous input by users. Make sure that functions that are triggered by moving a device can also be operated by conventional user interface components”	Harvard: https://accessibility.huit.harvard.edu/design-controls-easy-operation
Tests students motor skills	The device is vertical and challenges the motor skills of the students in a way that few devices in the classroom do. It requires the students to work on a different range of motion.	Yes	“Playing on vertical surfaces can also promote mobility in kids arms and shoulders, placing objects high enough so kids have to reach to grab is a great way to children to stretch and also give them a chance to practice their balancing, another gross motor skill.”	Tiny Feats: https://tinyfeats.com/blogs/news/the-value-in-vertical-play
Number of tiles	We will have four tiles on the opposite sides of the device that can be used by one or two students.	Yes	While interviewing Mrs. Brennan, we were told that a lot of the matching activities the students do is only with one or two matches, and it focuses on them explaining the matches. For this reason we will only have two matches on each side.	Interview with Mrs. Brennan